Flutter

Complete Developer Guide

From Zero to Production

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#1 Introduction to Flutter

Flutter is Google's open-source UI toolkit for building natively compiled, beautiful applications for mobile (Android & iOS), web, and desktop from a single Dart codebase. Flutter uses the Dart language and its own rendering engine (Skia/Impeller) — it does NOT rely on platform WebViews or OEM widgets.

- Hot Reload — See changes instantly without losing app state
- Rich widget catalog — Material Design & Cupertino widgets built-in
- Single codebase for Android, iOS, Web, Windows, macOS, Linux
- High performance — compiled to native ARM or x86 machine code
- Strong ecosystem — pub.dev has 35,000+ packages

■ *flutter create my_app — creates a new Flutter project*

```dart
// main.dart — Minimal Flutter app
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      home: const MyHomePage(title: 'Welcome to Flutter'),
    );
  }
}

class MyHomePage extends StatelessWidget {
  final String title;
  const MyHomePage({super.key, required this.title});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(title)),
      body: const Center(
        child: Text(
          'Hello, Flutter! ■',
          style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
        ),
      ),
    );
  }
}
```

#2 Variables — String, int, double, bool

Dart is a statically typed language. Every variable has a type. The four most common primitive types are String, int, double, and bool.

```
void main() {
  // ■■ String ■■■
```

```
  var city = 'Lagos';          // inferred as String
  var count = 10;              // inferred as int
  var discount = 0.15;         // inferred as double
  // city = 42;               // ■ Error: int is not a String

  print(city.toUpperCase());   // LAGOS

  // ■■ dynamic ■■■
```

```
  // ■■ const ■■■
```

```
  fruits.removeLast();
  print(fruits);

  // ■■ Iteration ■■■
```

```
  print(ages['Abdul']);           // 25
  print(ages['Unknown']);         // null
  print(ages['Bello'] ?? 0);      // 22 (null-safe access)
  print(user['name']);            // Abdul

  // ■■ Adding & updating ■■■
```

```
    print('Grade: B');
  } else if (score >= 70) {
    print('Grade: C');
  } else if (score >= 60) {
    print('Grade: D');
  } else {
    print('Grade: F');
  }

  // ■■ Ternary operator ■■■
```

Dart provides for, for-in, while, do-while loops, plus forEach and list comprehension-style patterns.

```dart
void main() {
  // ■■ for loop ■■■
```

#9 Classes

Dart is an object-oriented language. Classes encapsulate data (fields) and behaviour (methods). Dart supports constructors, named constructors, factory constructors, getters/setters, and more.

```dart
// ■■ Basic Class ■■■
```

#10 Inheritance

Dart supports single inheritance using the extends keyword. Subclasses inherit all non-private members and can override methods. Mixins allow code reuse across multiple class hierarchies.

```
// ■■ Base class ■■■
```

```
  double radius;
  Circle(this.radius);
  @override double get area => 3.14159 * radius * radius;
  @override double get perimeter => 2 * 3.14159 * radius;
}

// ■■ Mixin ■■■
```

```
        padding: const EdgeInsets.all(16),
        child: Text(
          'Hello, \$name! ■',
          style: const TextStyle(fontSize: 18, color: Colors.white),
        ),
      ),
    );
  }
}

// ■■ StatefulWidget — has mutable state ■■■
```

```dart
import 'package:flutter/material.dart';

class TextExamples extends StatelessWidget {
  const TextExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Text Widget')),
      body: const Padding(
        padding: EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // Basic text
            Text('Hello, Flutter!'),

            SizedBox(height: 12),

            // Styled text
            Text(
              'Bold & Large',
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
                color: Colors.deepPurple,
                letterSpacing: 1.5,
              ),
            ),

            // Italic & colored
            Text(
              'Styled Text Example',
              style: TextStyle(
                fontStyle: FontStyle.italic,
                color: Colors.teal,
                decoration: TextDecoration.underline,
                decorationColor: Colors.teal,
              ),
            ),

            SizedBox(height: 8),

            // Overflow handling
            Text(
              'This is a very long text that will overflow if not handled properly...',
              maxLines: 2,
              overflow: TextOverflow.ellipsis,
              style: TextStyle(fontSize: 14),
            ),

            SizedBox(height: 8),

            // Text alignment
            SizedBox(
              width: double.infinity,
              child: Text(
                'Centered Text',
                textAlign: TextAlign.center,
                style: TextStyle(fontSize: 18),
              ),
```

```
            ),

            // Text with shadow
            Text(
              'Shadow Effect',
              style: TextStyle(
                fontSize: 22,
                fontWeight: FontWeight.bold,
                shadows: [
                  Shadow(color: Colors.black45, blurRadius: 4, offset: Offset(2, 2)),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

#13 Container Widget

```
import 'package:flutter/material.dart';

class ContainerExamples extends StatelessWidget {
  const ContainerExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: SingleChildScrollView(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            // Basic container
            Container(
              width: 200,
              height: 100,
              color: Colors.blue,
              child: const Center(child: Text('Basic', style: TextStyle(color: Colors.white))),
            ),

            const SizedBox(height: 16),

            // Decorated container
            Container(
              width: double.infinity,
              height: 120,
              decoration: BoxDecoration(
                gradient: const LinearGradient(
                  colors: [Colors.purple, Colors.deepPurple],
                  begin: Alignment.topLeft,
                  end: Alignment.bottomRight,
                ),
                borderRadius: BorderRadius.circular(16),
                boxShadow: const [
                  BoxShadow(color: Colors.black26, blurRadius: 8, offset: Offset(0, 4)),
```

```
                ],
              ),
              child: const Center(
                child: Text('Gradient Container', style: TextStyle(color: Colors.white, fontSize: 20)),
              ),
            ),

            const SizedBox(height: 16),

            // Container with border
            Container(
              padding: const EdgeInsets.all(16),
              margin: const EdgeInsets.symmetric(vertical: 8),
              decoration: BoxDecoration(
                color: Colors.white,
                border: Border.all(color: Colors.deepPurple, width: 2),
                borderRadius: BorderRadius.circular(12),
              ),
              child: const Text('Bordered Container', style: TextStyle(fontSize: 16)),
            ),

            // Circular container (avatar-like)
            Container(
              width: 80,
              height: 80,
              decoration: const BoxDecoration(
                color: Colors.orange,
                shape: BoxShape.circle,
              ),
              child: const Icon(Icons.person, color: Colors.white, size: 40),
            ),
          ],
        ),
      ),
    );
  }
}
```

#14 Row & Column Widget

```
import 'package:flutter/material.dart';

class RowColumnExample extends StatelessWidget {
  const RowColumnExample({super.key});

  Widget _box(Color color, String label) {
    return Container(
      width: 60, height: 60, color: color,
      child: Center(child: Text(label, style: const TextStyle(color: Colors.white, fontSize: 11))),
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Padding(
        padding: const EdgeInsets.all(16),
```

```
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          const Text('Row — MainAxisAlignment.spaceEvenly'),
          Row(
            mainAxisAlignment: MainAxisAlignment.spaceEvenly,
            children: [_box(Colors.red, 'R'), _box(Colors.green, 'G'), _box(Colors.blue, 'B')],
          ),

          const SizedBox(height: 16),

          const Text('Row — CrossAxisAlignment.end'),
          Row(
            crossAxisAlignment: CrossAxisAlignment.end,
            children: [
              _box(Colors.purple, 'A'),
              Container(width: 60, height: 90, color: Colors.orange,
                  child: const Center(child: Text('B', style: TextStyle(color: Colors.white)))),
              _box(Colors.teal, 'C'),
            ],
          ),

          const SizedBox(height: 16),

          const Text('Expanded in Row'),
          Row(
            children: [
              _box(Colors.red, '1'),
              Expanded(child: Container(height: 60, color: Colors.amber,
                  child: const Center(child: Text('Expanded')))),
              _box(Colors.blue, '3'),
            ],
          ),

          const SizedBox(height: 16),

          const Text('Column — MainAxisAlignment.center'),
          SizedBox(
            height: 200,
            child: Column(
              mainAxisAlignment: MainAxisAlignment.spaceAround,
              crossAxisAlignment: CrossAxisAlignment.center,
              children: [
                _box(Colors.cyan, 'Top'),
                _box(Colors.indigo, 'Mid'),
                _box(Colors.pink, 'Bot'),
              ],
            ),
          ),

          // Flexible vs Expanded
          Row(
            children: [
              Flexible(flex: 1, child: Container(height: 40, color: Colors.red)),
              Flexible(flex: 2, child: Container(height: 40, color: Colors.green)),
              Flexible(flex: 1, child: Container(height: 40, color: Colors.blue)),
            ],
          ),
        ],
      ),
    ),
```

```
    );
  }
}
```

#15 Stack & Positioned Widget

```
import 'package:flutter/material.dart';

class StackExample extends StatelessWidget {
  const StackExample({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            // ■■ Profile card with Stack ■■■■■■■■■■■■■■■■■■■■■■■■■■■■
            Stack(
              children: [
                // Background image / gradient
                Container(
                  height: 200,
                  decoration: BoxDecoration(
                    borderRadius: BorderRadius.circular(16),
                    gradient: const LinearGradient(
                      colors: [Colors.deepPurple, Colors.blue],
                      begin: Alignment.topLeft,
                      end: Alignment.bottomRight,
                    ),
                  ),
                ),
                // Avatar positioned at top-center
                const Positioned(
                  top: 30,
                  left: 0,
                  right: 0,
                  child: CircleAvatar(
                    radius: 40,
                    backgroundColor: Colors.white,
                    child: Icon(Icons.person, size: 50, color: Colors.deepPurple),
                  ),
                ),
                // Name at bottom
                const Positioned(
                  bottom: 20,
                  left: 0,
                  right: 0,
                  child: Text(
                    'Abdul Dev',
                    textAlign: TextAlign.center,
                    style: TextStyle(color: Colors.white, fontSize: 20, fontWeight: FontWeight.bold),
                  ),
                ),
                // Badge top-right
                Positioned(
```

```
                  top: 12,
                  right: 12,
                  child: Container(
                    padding: const EdgeInsets.symmetric(horizontal: 8, vertical: 4),
                    decoration: BoxDecoration(color: Colors.amber, borderRadius: BorderRadius.circular(20)),
                    child: const Text('PRO', style: TextStyle(fontWeight: FontWeight.bold, fontSize: 12)),
                  ),
                ),
              ],
            ),

            const SizedBox(height: 24),

            // ■■ Alignment-based Stack ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
            Stack(
              alignment: Alignment.center,
              children: [
                Container(width: 150, height: 150, color: Colors.blue.shade100),
                Container(width: 100, height: 100, color: Colors.blue.shade300),
                Container(width: 50,  height: 50,  color: Colors.blue.shade600),
              ],
            ),
          ],
        ),
      ),
    );
  }
}
```

#16 Image Widget

```
import 'package:flutter/material.dart';

class ImageExamples extends StatelessWidget {
  const ImageExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: SingleChildScrollView(
        padding: const EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // ■■ Network image ■■■■■■■■■■■■■■■■■■■
```

```
const SizedBox(height: 16),

// ■■ Asset image ■■■
```

```
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

#17 Types of Buttons

```
import 'package:flutter/material.dart';

class ButtonExamples extends StatelessWidget {
  const ButtonExamples({super.key});

  void _show(BuildContext ctx, String name) {
    ScaffoldMessenger.of(ctx).showSnackBar(SnackBar(content: Text('\$name tapped')));
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.stretch,
          children: [
            // ElevatedButton
            ElevatedButton(
              onPressed: () => _show(context, 'ElevatedButton'),
              style: ElevatedButton.styleFrom(
                backgroundColor: Colors.deepPurple,
                foregroundColor: Colors.white,
                padding: const EdgeInsets.symmetric(vertical: 14),
                shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
              ),
              child: const Text('ElevatedButton', style: TextStyle(fontSize: 16)),
            ),

            const SizedBox(height: 12),

            // TextButton
            TextButton(
              onPressed: () => _show(context, 'TextButton'),
              child: const Text('TextButton', style: TextStyle(color: Colors.deepPurple, fontSize: 16)),
            ),

            // OutlinedButton
            OutlinedButton(
              onPressed: () => _show(context, 'OutlinedButton'),
              style: OutlinedButton.styleFrom(
                side: const BorderSide(color: Colors.deepPurple, width: 2),
                shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
              ),
              child: const Text('OutlinedButton'),
            ),
```

```
            const SizedBox(height: 12),

            // IconButton
            Row(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                IconButton(icon: const Icon(Icons.favorite, color: Colors.red),   onPressed: () {}),
                IconButton(icon: const Icon(Icons.share,    color: Colors.blue),   onPressed: () {}),
                IconButton(icon: const Icon(Icons.bookmark, color: Colors.green),  onPressed: () {}),
              ],
            ),

            // FloatingActionButton
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceEvenly,
              children: [
                FloatingActionButton(
                  heroTag: 'fab1',
                  onPressed: () {},
                  child: const Icon(Icons.add),
                ),
                FloatingActionButton.extended(
                  heroTag: 'fab2',
                  onPressed: () {},
                  label: const Text('Buy Now'),
                  icon: const Icon(Icons.shopping_cart),
                  backgroundColor: Colors.orange,
                ),
              ],
            ),

            const SizedBox(height: 12),

            // Custom gradient button using GestureDetector
            GestureDetector(
              onTap: () => _show(context, 'Custom Button'),
              child: Container(
                padding: const EdgeInsets.symmetric(vertical: 16),
                decoration: BoxDecoration(
                  gradient: const LinearGradient(colors: [Colors.purple, Colors.blue]),
                  borderRadius: BorderRadius.circular(12),
                ),
                child: const Center(
                  child: Text('Custom Gradient Button',
                      style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold)),
                ),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

#18 TextField & TextFormField

```dart
import 'package:flutter/material.dart';

class TextFieldExample extends StatefulWidget {
  const TextFieldExample({super.key});
  @override
  State<TextFieldExample> createState() => _TextFieldExampleState();
}

class _TextFieldExampleState extends State<TextFieldExample> {
  final _nameController    = TextEditingController();
  final _emailController   = TextEditingController();
  final _passwordController = TextEditingController();
  bool _obscure = true;

  @override
  void dispose() {
    _nameController.dispose();
    _emailController.dispose();
    _passwordController.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            // Basic TextField
            TextField(
              controller: _nameController,
              decoration: InputDecoration(
                labelText: 'Full Name',
                hintText: 'Enter your name',
                prefixIcon: const Icon(Icons.person),
                border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
                filled: true,
                fillColor: Colors.grey.shade100,
              ),
              textCapitalization: TextCapitalization.words,
              onChanged: (val) => print('Name: \$val'),
            ),

            const SizedBox(height: 16),

            // Email TextField
            TextField(
              controller: _emailController,
              keyboardType: TextInputType.emailAddress,
              decoration: InputDecoration(
                labelText: 'Email Address',
                prefixIcon: const Icon(Icons.email),
                border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
              ),
            ),

            const SizedBox(height: 16),

            // Password TextField with toggle
            TextField(
```

```
              controller: _passwordController,
              obscureText: _obscure,
              decoration: InputDecoration(
                labelText: 'Password',
                prefixIcon: const Icon(Icons.lock),
                suffixIcon: IconButton(
                  icon: Icon(_obscure ? Icons.visibility : Icons.visibility_off),
                  onPressed: () => setState(() => _obscure = !_obscure),
                ),
                border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
              ),
            ),

            const SizedBox(height: 16),

            // Multiline TextField
            const TextField(
              maxLines: 4,
              decoration: InputDecoration(
                labelText: 'Bio / Notes',
                alignLabelWithHint: true,
                border: OutlineInputBorder(),
                hintText: 'Write something about yourself...',
              ),
            ),

            const SizedBox(height: 16),

            ElevatedButton(
              onPressed: () {
                print('Name: \${_nameController.text}');
                print('Email: \${_emailController.text}');
              },
              child: const Text('Submit'),
            ),
          ],
        ),
      ),
    );
  }
}
```

#19 Form Validation

```
import 'package:flutter/material.dart';

class LoginForm extends StatefulWidget {
  const LoginForm({super.key});
  @override
  State<LoginForm> createState() => _LoginFormState();
}

class _LoginFormState extends State<LoginForm> {
  final _formKey = GlobalKey<FormState>();
  final _emailCtrl = TextEditingController();
  final _passCtrl  = TextEditingController();
  bool _loading = false;
```

```dart
String? _validateEmail(String? val) {
  if (val == null || val.isEmpty) return 'Email is required';
  final regex = RegExp(r'^[\w\.-]+@[\w\.-]+\.\w{2,}$');
  if (!regex.hasMatch(val)) return 'Enter a valid email address';
  return null;
}

String? _validatePassword(String? val) {
  if (val == null || val.isEmpty) return 'Password is required';
  if (val.length < 6) return 'Minimum 6 characters';
  if (!val.contains(RegExp(r'[A-Z]'))) return 'Include at least one uppercase letter';
  return null;
}

Future<void> _submit() async {
  if (_formKey.currentState!.validate()) {
    setState(() => _loading = true);
    await Future.delayed(const Duration(seconds: 2)); // simulate API
    setState(() => _loading = false);
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text('Login successful! ■'), backgroundColor: Colors.green),
    );
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Login')),
    body: Padding(
      padding: const EdgeInsets.all(24),
      child: Form(
        key: _formKey,
        child: Column(
          children: [
            TextFormField(
              controller: _emailCtrl,
              validator: _validateEmail,
              keyboardType: TextInputType.emailAddress,
              decoration: InputDecoration(
                labelText: 'Email',
                prefixIcon: const Icon(Icons.email),
                border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
              ),
            ),
            const SizedBox(height: 16),
            TextFormField(
              controller: _passCtrl,
              validator: _validatePassword,
              obscureText: true,
              decoration: InputDecoration(
                labelText: 'Password',
                prefixIcon: const Icon(Icons.lock),
                border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
              ),
            ),
            const SizedBox(height: 24),
            SizedBox(
              width: double.infinity,
              height: 50,
```

```
                child: ElevatedButton(
                  onPressed: _loading ? null : _submit,
                  child: _loading
                      ? const CircularProgressIndicator(color: Colors.white)
                      : const Text('Login', style: TextStyle(fontSize: 16)),
                ),
              ),
              TextButton(
                onPressed: () => _formKey.currentState!.reset(),
                child: const Text('Reset Form'),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

#20 ListView Widget

```
import 'package:flutter/material.dart';

class ListViewExamples extends StatelessWidget {
  final List<Map<String, dynamic>> contacts = const [
    {'name': 'Abdul',   'phone': '+234 801 000 0001', 'avatar': Icons.person},
    {'name': 'Amara',   'phone': '+234 802 000 0002', 'avatar': Icons.woman},
    {'name': 'Bello',   'phone': '+234 803 000 0003', 'avatar': Icons.person},
    {'name': 'Chioma',  'phone': '+234 804 000 0004', 'avatar': Icons.woman},
    {'name': 'Dayo',    'phone': '+234 805 000 0005', 'avatar': Icons.person},
  ];

  const ListViewExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return DefaultTabController(
      length: 3,
      child: Scaffold(
        appBar: AppBar(
          title: const Text('ListView Examples'),
          bottom: const TabBar(tabs: [
            Tab(text: 'Basic'), Tab(text: 'Builder'), Tab(text: 'Separated'),
          ]),
        ),
        body: TabBarView(
          children: [
            // ■■ ListView (direct children) ■■■■■■■■■■■■■■■■■■■■■■■■■
            ListView(
              padding: const EdgeInsets.all(8),
              children: contacts.map((c) => ListTile(
                leading: CircleAvatar(child: Icon(c['avatar'])),
                title: Text(c['name']),
                subtitle: Text(c['phone']),
                trailing: const Icon(Icons.call, color: Colors.green),
                onTap: () {},
              )).toList(),
```

```
            ),

            // ■■ ListView.builder (lazy, best for large lists) ■■■■■■■
            ListView.builder(
              itemCount: contacts.length,
              itemBuilder: (context, index) {
                final c = contacts[index];
                return Card(
                  margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 6),
                  child: ListTile(
                    leading: CircleAvatar(
                      backgroundColor: Colors.deepPurple,
                      child: Text(c['name'][0], style: const TextStyle(color: Colors.white)),
                    ),
                    title: Text(c['name'], style: const TextStyle(fontWeight: FontWeight.bold)),
                    subtitle: Text(c['phone']),
                    trailing: IconButton(icon: const Icon(Icons.delete, color: Colors.red), onPressed: () {}),
                  ),
                );
              },
            ),

            // ■■ ListView.separated ■■■■■■■
```

```
  factory Post.fromJson(Map<String, dynamic> json) => Post(
        id: json['id'],
        title: json['title'],
        body: json['body'],
      );
}

// ■■ API Service ■■■
```

```
          }
          final posts = snapshot.data!;
          return ListView.builder(
            itemCount: posts.length,
            itemBuilder: (_, i) => Card(
              margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 6),
              child: ListTile(
                leading: CircleAvatar(child: Text('\${posts[i].id}')),
                title: Text(posts[i].title, maxLines: 1, overflow: TextOverflow.ellipsis,
                    style: const TextStyle(fontWeight: FontWeight.bold)),
                subtitle: Text(posts[i].body, maxLines: 2, overflow: TextOverflow.ellipsis),
              ),
            ),
          );
        },
      ),
    );
  }
}
```

#22 GridView Widget

```
import 'package:flutter/material.dart';

class GridViewExamples extends StatelessWidget {
  const GridViewExamples({super.key});

  @override
  Widget build(BuildContext context) {
    final items = List.generate(20, (i) => 'Item \${i + 1}');
    final colors = [Colors.red, Colors.blue, Colors.green, Colors.orange,
                    Colors.purple, Colors.teal, Colors.pink, Colors.indigo];

    return DefaultTabController(
      length: 2,
      child: Scaffold(
        appBar: AppBar(
          title: const Text('GridView'),
          bottom: const TabBar(tabs: [Tab(text: 'Count'), Tab(text: 'Extent')]),
        ),
        body: TabBarView(
          children: [
            // ■■ GridView.count ■■■
```

```
                children: [
                  Icon(Icons.star, color: Colors.white, size: 28),
                  const SizedBox(height: 4),
                  Text(item, style: const TextStyle(color: Colors.white, fontSize: 11)),
                ],
              ),
            );
          }).toList(),
        ),

        // ■■ GridView.builder with extent ■■■■■■■■■■■■■■■■■■■■■■■■
        GridView.builder(
          padding: const EdgeInsets.all(12),
          gridDelegate: const SliverGridDelegateWithMaxCrossAxisExtent(
            maxCrossAxisExtent: 160,   // max width per cell
            childAspectRatio: 0.75,    // height/width ratio
            crossAxisSpacing: 8,
            mainAxisSpacing: 8,
          ),
          itemCount: 20,
          itemBuilder: (context, index) {
            return Card(
              clipBehavior: Clip.hardEdge,
              shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
              child: Column(
                children: [
                  Expanded(
                    child: Container(
                      color: colors[index % colors.length],
                      child: const Center(child: Icon(Icons.image, color: Colors.white, size: 40)),
                    ),
                  ),
                  Padding(
                    padding: const EdgeInsets.all(8),
                    child: Text('Product \${index + 1}', style: const TextStyle(fontWeight: FontWeight.bold)),
                  ),
                ],
              ),
            );
          },
        ),
      ],
    ),
  ),
);
  }
}
```

#23 Navigation

```
import 'package:flutter/material.dart';

// ■■ Screen A ■■■■■■■■■■■■■■■■■■■
```

```
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Screen A')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            // Push — add to stack
            ElevatedButton(
              onPressed: () => Navigator.push(
                context,
                MaterialPageRoute(builder: (_) => const ScreenB(title: 'From A')),
              ),
              child: const Text('Go to Screen B'),
            ),

            // Push and wait for result
            ElevatedButton(
              onPressed: () async {
                final result = await Navigator.push<String>(
                  context,
                  MaterialPageRoute(builder: (_) => const PickerScreen()),
                );
                if (result != null) print('Got: \$result');
              },
              child: const Text('Open Picker'),
            ),

            // Named routes (configure in MaterialApp)
            ElevatedButton(
              onPressed: () => Navigator.pushNamed(context, '/details', arguments: {'id': 42}),
              child: const Text('Named Route'),
            ),

            // Push and remove all previous
            ElevatedButton(
              onPressed: () => Navigator.pushAndRemoveUntil(
                context,
                MaterialPageRoute(builder: (_) => const ScreenA()),
                (route) => false,           // remove all
              ),
              child: const Text('Push & Clear Stack'),
            ),
          ],
        ),
      ),
    );
  }
}

// ■■ Screen B ■■■
```

```
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Title received: \$title'),
            ElevatedButton(
              onPressed: () => Navigator.pop(context),
              child: const Text('← Go Back'),
            ),
            ElevatedButton(
              onPressed: () => Navigator.pushReplacement(
                context,
                MaterialPageRoute(builder: (_) => const ScreenA()),
              ),
              child: const Text('Replace with A'),
            ),
          ],
        ),
      ),
    );
  }
}

// ■■ Picker Screen — returns a value ■■■
```

```
    this.color = Colors.deepPurple,
    this.isLoading = false,
  });

  @override
  Widget build(BuildContext context) {
    return SizedBox(
      width: double.infinity,
      height: 52,
      child: ElevatedButton(
        onPressed: isLoading ? null : onPressed,
        style: ElevatedButton.styleFrom(
          backgroundColor: color,
          foregroundColor: Colors.white,
          shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(14)),
        ),
        child: isLoading
            ? const SizedBox(width: 20, height: 20,
                child: CircularProgressIndicator(color: Colors.white, strokeWidth: 2))
            : Row(
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                  if (icon != null) ...[Icon(icon), const SizedBox(width: 8)],
                  Text(label, style: const TextStyle(fontSize: 16, fontWeight: FontWeight.bold)),
                ],
              ),
      ),
    );
  }
}

// ■■ Reusable info card ■■■
```

```
            Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(title, style: TextStyle(color: Colors.grey.shade600, fontSize: 12)),
                Text(value, style: const TextStyle(fontWeight: FontWeight.bold, fontSize: 20)),
              ],
            ),
          ],
        ),
      ),
    );
  }
}

// ■■ Usage ■■■
```

```
        ),
        ElevatedButton(
          style: ElevatedButton.styleFrom(backgroundColor: Colors.red),
          onPressed: () { Navigator.pop(context); /* delete */ },
          child: const Text('Delete', style: TextStyle(color: Colors.white)),
        ),
      ],
    ),
  );
}

// ■■ Custom AlertDialog ■■■
```

```
      builder: (context) => AlertDialog(
        title: const Text('Enter Name'),
        content: TextField(controller: ctrl, decoration: const InputDecoration(hintText: 'Your name')),
        actions: [
          TextButton(onPressed: () => Navigator.pop(context), child: const Text('Cancel')),
          ElevatedButton(
            onPressed: () { print(ctrl.text); Navigator.pop(context); },
            child: const Text('OK'),
          ),
        ],
      ),
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            ElevatedButton(onPressed: () => _showBasic(context),  child: const Text('Basic Dialog')),
            ElevatedButton(onPressed: () => _showCustom(context), child: const Text('Custom Dialog')),
            ElevatedButton(onPressed: () => _showInput(context),  child: const Text('Input Dialog')),
          ],
        ),
      ),
    );
  }
}
```

#26 SnackBar

```
import 'package:flutter/material.dart';

class SnackBarExamples extends StatelessWidget {
  const SnackBarExamples({super.key});

  void _basic(BuildContext ctx) {
    ScaffoldMessenger.of(ctx).showSnackBar(
      const SnackBar(content: Text('This is a basic SnackBar')),
    );
  }

  void _withAction(BuildContext ctx) {
    ScaffoldMessenger.of(ctx).showSnackBar(
      SnackBar(
        content: const Text('Item deleted'),
        duration: const Duration(seconds: 4),
        action: SnackBarAction(
          label: 'UNDO',
          textColor: Colors.yellow,
          onPressed: () => print('Undo pressed'),
        ),
      ),
    );
  }
```

```
  void _styled(BuildContext ctx) {
    ScaffoldMessenger.of(ctx).showSnackBar(
      SnackBar(
        content: const Row(
          children: [
            Icon(Icons.check_circle, color: Colors.white),
            SizedBox(width: 12),
            Text('Payment successful! ■', style: TextStyle(color: Colors.white)),
          ],
        ),
        backgroundColor: Colors.green.shade700,
        behavior: SnackBarBehavior.floating,         // floats above bottom
        shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
        margin: const EdgeInsets.all(16),
        duration: const Duration(seconds: 3),
      ),
    );
  }

  void _error(BuildContext ctx) {
    ScaffoldMessenger.of(ctx).showSnackBar(
      SnackBar(
        content: const Row(
          children: [
            Icon(Icons.error, color: Colors.white),
            SizedBox(width: 12),
            Text('Something went wrong!', style: TextStyle(color: Colors.white)),
          ],
        ),
        backgroundColor: Colors.red.shade700,
        behavior: SnackBarBehavior.floating,
        shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
        margin: const EdgeInsets.all(16),
      ),
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            ElevatedButton(onPressed: () => _basic(context),      child: const Text('Basic SnackBar')),
            ElevatedButton(onPressed: () => _withAction(context), child: const Text('With Action')),
            ElevatedButton(onPressed: () => _styled(context),     child: const Text('Success SnackBar')),
            ElevatedButton(onPressed: () => _error(context),      child: const Text('Error SnackBar')),
          ],
        ),
      ),
    );
  }
}
```

#27 Navigation Drawer

```dart
import 'package:flutter/material.dart';

class DrawerScreen extends StatefulWidget {
  const DrawerScreen({super.key});
  @override
  State<DrawerScreen> createState() => _DrawerScreenState();
}

class _DrawerScreenState extends State<DrawerScreen> {
  int _selectedIndex = 0;

  final _pages = const [
    Center(child: Text('■ Home Page', style: TextStyle(fontSize: 24))),
    Center(child: Text('■ Profile Page', style: TextStyle(fontSize: 24))),
    Center(child: Text('■■ Settings Page', style: TextStyle(fontSize: 24))),
  ];

  final _titles = ['Home', 'Profile', 'Settings'];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_titles[_selectedIndex])),
      drawer: Drawer(
        child: ListView(
          padding: EdgeInsets.zero,
          children: [
            // Header
            UserAccountsDrawerHeader(
              accountName: const Text('Abdul Developer', style: TextStyle(fontWeight: FontWeight.bold)),
              accountEmail: const Text('abdul@yenko.ng'),
              currentAccountPicture: const CircleAvatar(
                backgroundColor: Colors.white,
                child: Icon(Icons.person, color: Colors.deepPurple, size: 40),
              ),
              decoration: const BoxDecoration(
                gradient: LinearGradient(colors: [Colors.deepPurple, Colors.blue]),
              ),
            ),

            // Menu items
            _drawerItem(Icons.home,     'Home',     0),
            _drawerItem(Icons.person,   'Profile',  1),
            _drawerItem(Icons.settings, 'Settings', 2),

            const Divider(),

            ListTile(
              leading: const Icon(Icons.help_outline),
              title: const Text('Help & Support'),
              onTap: () { Navigator.pop(context); },
            ),

            ListTile(
              leading: const Icon(Icons.logout, color: Colors.red),
              title: const Text('Logout', style: TextStyle(color: Colors.red)),
              onTap: () { Navigator.pop(context); },
            ),
          ],
        ),
      ),
```

```
      body: _pages[_selectedIndex],
    );
  }

  ListTile _drawerItem(IconData icon, String label, int index) {
    return ListTile(
      leading: Icon(icon, color: _selectedIndex == index ? Colors.deepPurple : null),
      title: Text(label, style: TextStyle(
        fontWeight: _selectedIndex == index ? FontWeight.bold : FontWeight.normal,
        color: _selectedIndex == index ? Colors.deepPurple : null,
      )),
      selected: _selectedIndex == index,
      onTap: () {
        setState(() => _selectedIndex = index);
        Navigator.pop(context);
      },
    );
  }
}
```

#28 AppBar Widget

```
import 'package:flutter/material.dart';

class AppBarExamples extends StatelessWidget {
  const AppBarExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      // ■■ Standard AppBar ■■■
```

```
// ■■ SliverAppBar (collapsible) ■■■
```

```
          backgroundColor: Colors.deepPurple,
          bottom: const TabBar(
            isScrollable: false,
            indicatorColor: Colors.amber,
            labelColor: Colors.white,
            unselectedLabelColor: Colors.white60,
            tabs: [
              Tab(icon: Icon(Icons.home),        text: 'Home'),
              Tab(icon: Icon(Icons.explore),     text: 'Explore'),
              Tab(icon: Icon(Icons.favorite),    text: 'Saved'),
              Tab(icon: Icon(Icons.person),      text: 'Profile'),
            ],
          ),
        ),
        body: const TabBarView(
          children: [
            _TabContent(label: '■ Home', color: Colors.blue),
            _TabContent(label: '■ Explore', color: Colors.green),
            _TabContent(label: '❤■ Saved', color: Colors.red),
            _TabContent(label: '■ Profile', color: Colors.purple),
          ],
        ),
      ),
    );
  }
}

class _TabContent extends StatelessWidget {
  final String label;
  final Color color;
  const _TabContent({required this.label, required this.color});

  @override
  Widget build(BuildContext context) {
    return Container(
      color: color.withOpacity(0.05),
      child: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(Icons.widgets, color: color, size: 80),
            const SizedBox(height: 16),
            Text(label, style: TextStyle(fontSize: 28, color: color, fontWeight: FontWeight.bold)),
          ],
        ),
      ),
    );
  }
}
```

#30 BottomSheet Dialog

```
import 'package:flutter/material.dart';

class BottomSheetExamples extends StatelessWidget {
  const BottomSheetExamples({super.key});
```

```
void _showModal(BuildContext ctx) {
  showModalBottomSheet(
    context: ctx,
    shape: const RoundedRectangleBorder(
      borderRadius: BorderRadius.vertical(top: Radius.circular(24)),
    ),
    builder: (context) => DraggableScrollableSheet(
      initialChildSize: 0.4,
      minChildSize: 0.2,
      maxChildSize: 0.8,
      expand: false,
      builder: (_, controller) => Column(
        children: [
          Container(
            margin: const EdgeInsets.symmetric(vertical: 10),
            width: 40, height: 4,
            decoration: BoxDecoration(color: Colors.grey.shade300, borderRadius: BorderRadius.circular(4)),
          ),
          const Text('Choose an Option', style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold)),
          const Divider(),
          Expanded(
            child: ListView(
              controller: controller,
              children: [
                ListTile(leading: const Icon(Icons.camera_alt), title: const Text('Camera'),   onTap: () => Navigator.
                ListTile(leading: const Icon(Icons.photo),       title: const Text('Gallery'),  onTap: () => Navigator
                ListTile(leading: const Icon(Icons.file_upload), title: const Text('File'),     onTap: () => Navigator
                ListTile(leading: const Icon(Icons.delete, color: Colors.red), title: const Text('Delete', style: Text
              ],
            ),
          ),
        ],
      ),
    ),
  );
}

void _showPersistent(BuildContext ctx) {
  showBottomSheet(
    context: ctx,
    builder: (context) => Container(
      padding: const EdgeInsets.all(16),
      child: const Row(
        children: [
          Icon(Icons.info, color: Colors.blue),
          SizedBox(width: 12),
          Expanded(child: Text('This is a persistent bottom sheet.')),
        ],
      ),
    ),
  );
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          ElevatedButton(onPressed: () => _showModal(context),      child: const Text('Modal BottomSheet')),
```

```
            ElevatedButton(onPressed: () => _showPersistent(context), child: const Text('Persistent BottomSheet')),
          ],
        ),
      ),
    );
  }
}
```

#31 BottomNavigationBar

```
import 'package:flutter/material.dart';

class BottomNavExample extends StatefulWidget {
  const BottomNavExample({super.key});
  @override
  State<BottomNavExample> createState() => _BottomNavExampleState();
}

class _BottomNavExampleState extends State<BottomNavExample> {
  int _currentIndex = 0;

  final _pages = [
    const Center(child: Text('■ Home', style: TextStyle(fontSize: 28))),
    const Center(child: Text('■ Search', style: TextStyle(fontSize: 28))),
    const Center(child: Text('■ Add', style: TextStyle(fontSize: 28))),
    const Center(child: Text('■ Notifications', style: TextStyle(fontSize: 28))),
    const Center(child: Text('■ Profile', style: TextStyle(fontSize: 28))),
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Bottom Nav Demo')),
      body: AnimatedSwitcher(
        duration: const Duration(milliseconds: 300),
        child: _pages[_currentIndex],
      ),
      bottomNavigationBar: BottomNavigationBar(
        currentIndex: _currentIndex,
        onTap: (i) => setState(() => _currentIndex = i),
        type: BottomNavigationBarType.fixed,   // show all labels
        selectedItemColor: Colors.deepPurple,
        unselectedItemColor: Colors.grey,
        selectedLabelStyle: const TextStyle(fontWeight: FontWeight.bold),
        items: const [
          BottomNavigationBarItem(icon: Icon(Icons.home),         label: 'Home'),
          BottomNavigationBarItem(icon: Icon(Icons.search),       label: 'Search'),
          BottomNavigationBarItem(icon: Icon(Icons.add_circle, size: 36), label: 'Add'),
          BottomNavigationBarItem(icon: Icon(Icons.notifications), label: 'Alerts'),
          BottomNavigationBarItem(icon: Icon(Icons.person),        label: 'Profile'),
        ],
      ),
    );
  }
}
```

#32 RichText Widget

```
import 'package:flutter/material.dart';
import 'package:flutter/gestures.dart';

class RichTextExamples extends StatelessWidget {
  const RichTextExamples({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            // Multi-style text
            RichText(
              text: const TextSpan(
                style: TextStyle(fontSize: 16, color: Colors.black),
                children: [
                  TextSpan(text: 'Hello '),
                  TextSpan(text: 'Flutter', style: TextStyle(color: Colors.blue, fontWeight: FontWeight.bold)),
                  TextSpan(text: '! This is '),
                  TextSpan(text: 'RichText', style: TextStyle(color: Colors.purple, fontStyle: FontStyle.italic)),
                  TextSpan(text: ' widget.'),
                ],
              ),
            ),

            const SizedBox(height: 16),

            // Mixed sizes & colors
            RichText(
              text: const TextSpan(
                children: [
                  TextSpan(text: '■', style: TextStyle(fontSize: 18, color: Colors.green)),
                  TextSpan(text: '50,000', style: TextStyle(fontSize: 32, fontWeight: FontWeight.bold, color: Colors.gre
                  TextSpan(text: '.00', style: TextStyle(fontSize: 16, color: Colors.green)),
                ],
              ),
            ),

            const SizedBox(height: 16),

            // Tappable spans
            RichText(
              text: TextSpan(
                style: const TextStyle(fontSize: 14, color: Colors.black),
                children: [
                  const TextSpan(text: 'By signing up, you agree to our '),
                  TextSpan(
                    text: 'Terms of Service',
                    style: const TextStyle(color: Colors.blue, decoration: TextDecoration.underline),
                    recognizer: TapGestureRecognizer()..onTap = () => print('Terms tapped'),
                  ),
                  const TextSpan(text: ' and '),
                  TextSpan(
                    text: 'Privacy Policy',
                    style: const TextStyle(color: Colors.blue, decoration: TextDecoration.underline),
                    recognizer: TapGestureRecognizer()..onTap = () => print('Privacy tapped'),
                  ),
```

```
                  const TextSpan(text: '.'),
                ],
              ),
            ),

            const SizedBox(height: 16),

            // Inline code-like styling
            RichText(
              text: const TextSpan(
                style: TextStyle(fontSize: 14, color: Colors.black87),
                children: [
                  TextSpan(text: 'Use '),
                  TextSpan(text: 'setState()', style: TextStyle(
                    fontFamily: 'Courier',
                    backgroundColor: Color(0xFFEEEEEE),
                    color: Colors.red,
                    fontSize: 13,
                  )),
                  TextSpan(text: ' to trigger a rebuild in a '),
                  TextSpan(text: 'StatefulWidget', style: TextStyle(
                    fontFamily: 'Courier',
                    backgroundColor: Color(0xFFEEEEEE),
                    color: Colors.red,
                    fontSize: 13,
                  )),
                  TextSpan(text: '.'),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

#33 DropdownMenu Widget

```
import 'package:flutter/material.dart';

class DropdownExamples extends StatefulWidget {
  const DropdownExamples({super.key});
  @override
  State<DropdownExamples> createState() => _DropdownExamplesState();
}

class _DropdownExamplesState extends State<DropdownExamples> {
  String? _selectedCountry;
  String? _selectedNetwork;
  String? _selectedPlan;

  final countries = ['Nigeria', 'Ghana', 'Kenya', 'South Africa', 'Egypt'];
  final networks  = ['MTN', 'Airtel', 'Glo', '9Mobile'];
  final plans     = ['500MB - ■200', '1GB - ■350', '2GB - ■600', '5GB - ■1,200'];

  @override
```

```
Widget build(BuildContext context) {
  return Scaffold(
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          const Text('Select Country', style: TextStyle(fontWeight: FontWeight.bold)),
          const SizedBox(height: 8),

          // ■■ DropdownButton ■■■
```

```
            Text('Plan: \${_selectedPlan ?? "—"}'),
          ],
        ),
      ),
    );
  }
}
```

#34 Using External Packages

External packages are added via pub.dev. Add them to pubspec.yaml then run flutter pub get. Here are common Flutter packages with usage examples.

```
# pubspec.yaml
dependencies:
  flutter:
    sdk: flutter

  # Networking
  http: ^1.2.0
  dio: ^5.4.0

  # State management
  get: ^4.6.6
  provider: ^6.1.1

  # Local storage
  shared_preferences: ^2.2.2
  get_storage: ^2.1.1

  # UI & Design
  flutter_svg: ^2.0.9
  cached_network_image: ^3.3.1
  shimmer: ^3.0.0
  lottie: ^3.0.0

  # Utilities
  intl: ^0.19.0
  url_launcher: ^6.2.4
  image_picker: ^1.0.7
```

```
import 'package:flutter/material.dart';
import 'package:cached_network_image/cached_network_image.dart';
import 'package:shimmer/shimmer.dart';
import 'package:intl/intl.dart';
import 'package:url_launcher/url_launcher.dart';

// ■■ cached_network_image ■■■
```

```
      placeholder: (_, __) => Shimmer.fromColors(
        baseColor: Colors.grey.shade300,
        highlightColor: Colors.grey.shade100,
        child: Container(color: Colors.grey.shade300, height: 300),
      ),
      errorWidget: (_, __, ___) => const Icon(Icons.error),
      fit: BoxFit.cover,
    );
  }
}

// ■■ intl — formatting dates & currency ■■■
```

```
  final _box = GetStorage();

  // Keys
  static const _tokenKey   = 'auth_token';
  static const _themeKey   = 'is_dark_theme';
  static const _userKey    = 'user_data';
  static const _onboardKey = 'onboarding_done';

  // Auth token
  void saveToken(String token) => _box.write(_tokenKey, token);
  String? get token => _box.read<String>(_tokenKey);
  void clearToken() => _box.remove(_tokenKey);

  // Theme preference
  void setDarkMode(bool isDark) => _box.write(_themeKey, isDark);
  bool get isDarkMode => _box.read<bool>(_themeKey) ?? false;

  // User data (Map)
  void saveUser(Map<String, dynamic> user) => _box.write(_userKey, user);
  Map<String, dynamic>? get user => _box.read<Map<String, dynamic>>(_userKey);

  // Onboarding
  void markOnboardingDone() => _box.write(_onboardKey, true);
  bool get isOnboardingDone => _box.read<bool>(_onboardKey) ?? false;

  // Clear everything
  void clearAll() => _box.erase();
}

// ■■ Usage in widget ■■■
```

```
            leading: const Icon(Icons.delete, color: Colors.red),
            title: const Text('Clear All Storage', style: TextStyle(color: Colors.red)),
            onTap: () { _storage.clearAll(); setState(() {}); },
          ),
        ],
      ),
    );
  }
}
```

#36 Introduction to GetX

GetX is a powerful Flutter micro-framework that provides state management, dependency injection, and route management in one package. It is lightweight, fast, and highly productive.

- *flutter pub add get*

```
import 'package:flutter/material.dart';
import 'package:get/get.dart';

// ■■ 1. Replace MaterialApp with GetMaterialApp ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return GetMaterialApp(
      title: 'GetX Demo',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple)),
      // GetX named routes
      initialRoute: '/',
      getPages: [
        GetPage(name: '/',        page: () => const HomeScreen()),
        GetPage(name: '/profile', page: () => const ProfileScreen()),
        GetPage(name: '/detail',  page: () => const DetailScreen()),
      ],
    );
  }
}

// ■■ 2. GetX Snackbar, Dialog, BottomSheet ■■■■■■■■■■■■■■■■■■■
```

```
                backgroundColor: Colors.green,
                colorText: Colors.white,
                icon: const Icon(Icons.check, color: Colors.white),
                snackPosition: SnackPosition.BOTTOM,
              ),
              child: const Text('GetX Snackbar'),
            ),
            ElevatedButton(
              onPressed: () => Get.defaultDialog(
                title: 'Confirm',
                middleText: 'Are you sure?',
                onConfirm: () { Get.back(); },
                onCancel: () {},
              ),
              child: const Text('GetX Dialog'),
            ),
            ElevatedButton(
              onPressed: () => Get.to(() => const ProfileScreen()),
              child: const Text('Go to Profile'),
            ),
            ElevatedButton(
              onPressed: () => Get.toNamed('/profile'),
              child: const Text('Named Route'),
            ),
            ElevatedButton(
              onPressed: () => Get.offAllNamed('/'), // clear stack
              child: const Text('Off All & Go Home'),
            ),
          ],
        ),
      ),
    );
  }
}

class ProfileScreen extends StatelessWidget {
  const ProfileScreen({super.key});
  @override
  Widget build(BuildContext context) => Scaffold(
    appBar: AppBar(title: const Text('Profile')),
    body: Center(child: ElevatedButton(onPressed: () => Get.back(), child: const Text('Back'))),
  );
}

class DetailScreen extends StatelessWidget {
  const DetailScreen({super.key});
  @override
  Widget build(BuildContext context) => const Scaffold(body: Center(child: Text('Detail')));
}
```

#37 Passing Parameters Using GetX

```
import 'package:flutter/material.dart';
import 'package:get/get.dart';

// ■■ Model ■■■
```

```
  final int id;
  final String name;
  final double price;
  final String imageUrl;

  Product({required this.id, required this.name, required this.price, required this.imageUrl});
}

// ■■ List Screen ■■■
```

```
          children: [
            ClipRRect(
              borderRadius: BorderRadius.circular(16),
              child: Image.network(product.imageUrl, width: double.infinity, height: 200, fit: BoxFit.cover),
            ),
            const SizedBox(height: 16),
            Text(product.name, style: const TextStyle(fontSize: 24, fontWeight: FontWeight.bold)),
            const SizedBox(height: 8),
            Text('\$\${product.price.toStringAsFixed(2)}',
                style: const TextStyle(fontSize: 20, color: Colors.green)),
          ],
        ),
      ),
    );
  }
}

// ■■ Method 2: Parameters in Named Routes ■■■
```

```
  }

  // Increment / Decrement
  void increment(int index) {
    items[index]['qty']++;
    items.refresh();
    _recalculate();
  }

  void decrement(int index) {
    if (items[index]['qty'] > 1) {
      items[index]['qty']--;
      items.refresh();
    } else {
      items.removeAt(index);
    }
    _recalculate();
  }

  void _recalculate() {
    total.value = items.fold(0, (sum, i) => sum + i['price'] * i['qty']);
    itemCount.value = items.fold(0, (sum, i) => sum + (i['qty'] as int));
  }

  // Simulate API call
  Future<void> checkout() async {
    isLoading.value = true;
    await Future.delayed(const Duration(seconds: 2));
    items.clear();
    total.value = 0;
    itemCount.value = 0;
    isLoading.value = false;
    Get.snackbar('Success', 'Order placed! ■',
        backgroundColor: Colors.green, colorText: Colors.white);
  }

  @override
  void onInit() {
    super.onInit();
    // Called when controller is first created
    print('CartController initialized');
  }

  @override
  void onClose() {
    // Called when controller is removed from memory
    print('CartController disposed');
    super.onClose();
  }
}

// ■■ SHOP SCREEN ■■■
```

```
    {'name': 'GetX Mastery',   'price': 24.99},
    {'name': 'Widget Catalog', 'price': 14.99},
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Shop'),
        actions: [
          // Obx rebuilds ONLY this widget when itemCount changes
          Obx(() => Badge(
            label: Text('\${cart.itemCount.value}'),
            child: IconButton(
              icon: const Icon(Icons.shopping_cart),
              onPressed: () => Get.to(() => CartScreen()),
            ),
          )),
        ],
      ),
      body: GridView.builder(
        padding: const EdgeInsets.all(12),
        gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
          crossAxisCount: 2, crossAxisSpacing: 12, mainAxisSpacing: 12, childAspectRatio: 0.8,
        ),
        itemCount: products.length,
        itemBuilder: (_, i) {
          final p = products[i];
          return Card(
            shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
            child: Padding(
              padding: const EdgeInsets.all(12),
              child: Column(
                mainAxisAlignment: MainAxisAlignment.spaceBetween,
                children: [
                  const Icon(Icons.book, size: 60, color: Colors.deepPurple),
                  Text(p['name'] as String, textAlign: TextAlign.center,
                      style: const TextStyle(fontWeight: FontWeight.bold)),
                  Text('\$\${p['price']}', style: const TextStyle(color: Colors.green, fontSize: 16)),
                  ElevatedButton(
                    onPressed: () => cart.addItem(p['name'] as String, p['price'] as double),
                    child: const Text('Add to Cart'),
                  ),
                ],
              ),
            ),
          );
        },
      ),
    );
  }
}

// ■■ CART SCREEN ■■■
```

```
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Cart')),
    body: Obx(() {
      if (cart.isEmpty) {
        return const Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Icon(Icons.shopping_cart_outlined, size: 80, color: Colors.grey),
              SizedBox(height: 16),
              Text('Your cart is empty', style: TextStyle(fontSize: 18, color: Colors.grey)),
            ],
          ),
        );
      }
      return Column(
        children: [
          Expanded(
            child: ListView.builder(
              itemCount: cart.items.length,
              itemBuilder: (_, i) {
                final item = cart.items[i];
                return Card(
                  margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 6),
                  child: ListTile(
                    leading: const Icon(Icons.book, color: Colors.deepPurple, size: 36),
                    title: Text(item['name'], style: const TextStyle(fontWeight: FontWeight.bold)),
                    subtitle: Text('\$\${item['price']} × \${item['qty']}'),
                    trailing: Row(
                      mainAxisSize: MainAxisSize.min,
                      children: [
                        IconButton(icon: const Icon(Icons.remove_circle, color: Colors.red),
                            onPressed: () => cart.decrement(i)),
                        Text('\${item['qty']}', style: const TextStyle(fontWeight: FontWeight.bold, fontSize: 16)),
                        IconButton(icon: const Icon(Icons.add_circle, color: Colors.green),
                            onPressed: () => cart.increment(i)),
                      ],
                    ),
                  ),
                );
              },
            ),
          ),
          // Summary & Checkout
          Obx(() => Container(
            padding: const EdgeInsets.all(20),
            decoration: BoxDecoration(
              color: Colors.white,
              boxShadow: [BoxShadow(color: Colors.black12, blurRadius: 10, offset: const Offset(0, -4))],
              borderRadius: const BorderRadius.vertical(top: Radius.circular(24)),
            ),
            child: Column(
              children: [
                Row(
                  mainAxisAlignment: MainAxisAlignment.spaceBetween,
                  children: [
                    const Text('Total:', style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold)),
                    Text('\$\${cart.total.value.toStringAsFixed(2)}',
                        style: const TextStyle(fontSize: 22, fontWeight: FontWeight.bold, color: Colors.green)),
                  ],
```

```
                  ),
                  const SizedBox(height: 12),
                  SizedBox(
                    width: double.infinity,
                    height: 52,
                    child: ElevatedButton(
                      onPressed: cart.isLoading.value ? null : () => cart.checkout(),
                      style: ElevatedButton.styleFrom(
                        backgroundColor: Colors.deepPurple,
                        shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(14)),
                      ),
                      child: cart.isLoading.value
                          ? const CircularProgressIndicator(color: Colors.white)
                          : const Text('Checkout', style: TextStyle(fontSize: 18, color: Colors.white)),
                    ),
                  ),
                ],
              ),
            )),
          ],
        );
      }),
    );
  }
}
```

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